

AARON A

Full Stack Developer | MERN Stack Developer

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github.com/AaronAlen portfolio-pi-smoky-83.vercel.app LeetCode: 50+ Solved HackerRank: 30+ Solved

PROFESSIONAL SUMMARY

Full Stack Developer specializing in React.js, Node.js, Express.js, TypeScript, and MongoDB/MySQL with hands-on experience across two software engineering internships. Proficient in building modular frontend architectures, securing REST APIs with JWT & RBAC, and containerizing web services using Docker. Solved 50+ algorithmic problems on LeetCode (Blind 75) with solid fundamentals in Data Structures, Object-Oriented Programming (OOP), DBMS, and system optimization.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3, Python (Data Structures)
Frontend: React.js (v18/v19), Next.js, Redux Toolkit, Zustand, Context API, React Hooks, TanStack Query, TanStack Virtual, Tailwind CSS, Framer Motion, Responsive Design
Backend & APIs: Node.js, Express.js, REST API Architecture (Swagger / OpenAPI), WebSockets (Socket.IO), JWT Authentication, Role-Based Access Control (RBAC), Middleware Design
Databases: MongoDB, Mongoose ODM, MySQL 8 (InnoDB), Sequelize ORM, PostgreSQL (Supabase RLS), Aggregation Pipelines, Schema Normalization, Indexing & Query Optimization
DevOps & Tools: Docker, Docker Compose, Git, GitHub Actions (CI/CD), Postman, Linux CLI, Vercel, Render
AI & Integration: Groq AI API (LLaMA 3.3), LangGraph, Document Parsing (pdf-parse), Brevo REST API, Twilio SMS API, Nodemailer
Core Fundamentals & Testing: Data Structures & Algorithms (Blind 75), OOP, DBMS, Jest Unit Testing, API Integration Testing, Agile/Scrum

PROFESSIONAL EXPERIENCE

Software Engineer Trainee — GWC Data AI Jul 2026 – Oct 2026
ELCOT IT Park, Hosur, Tamil Nadu

- Engineered and deployed full-stack web applications using React.js, Node.js, Express, and PostgreSQL/MySQL, maintaining responsive data flow with sub-200ms API response times.
- Architected 25+ secure REST API endpoints with JWT dual-token authentication (15m access / 7d refresh tokens), bcrypt password hashing, input validation (Joi), and Role-Based Access Control (RBAC).
- Designed database schemas using Sequelize ORM and Mongoose; created optimized indexes and SQL joins that reduced query execution time by 35% across multi-table lookups.
- Built responsive user interfaces in React 18 with Redux Toolkit and Tailwind CSS, integrating Socket.IO for real-time bi-directional status streaming and state synchronization.
- Containerized multi-tier applications using Docker and Docker Compose; configured GitHub Actions CI/CD pipelines automating lint checks, test suites, and deployments on Render.
- Developed custom Domo Cloud analytical dashboards consuming internal REST APIs, implementing automated alerts and executive telemetry reporting.

MERN Stack Developer Intern — WHY Global Services Jan 2026 – Jul 2026
Chennai, Tamil Nadu, India

- Engineered and tested 5+ core RESTful API modules with Node.js, Express, and MongoDB covering user authentication, product catalogs, and order tracking.
- Built a modular UI library of 12+ reusable React/TypeScript components following clean component patterns, reducing redundant frontend development time by 30% across the engineering team.
- Implemented stateless JWT authentication and strict RBAC authorization middleware to secure API endpoints against unauthorized access.
- Constructed MongoDB aggregation pipelines (\$match, \$group, \$lookup, \$facet) for administrative inventory and sales reporting, reducing backend payload size by 40%.
- Authored 30+ automated unit and integration tests using Jest and Postman, achieving 85%+ test coverage across critical authentication and checkout routes.
- Collaborated in an Agile/Scrum workflow with daily stand-ups, two-week sprint planning, and GitHub PR reviews, utilizing Docker on Linux for consistent development environments.

PROJECTS

StockPilot — Autonomous AI Inventory Restock & Procurement System Live Demo | GitHub

React 18, TypeScript, Redux Toolkit, Node.js, Express.js (42 APIs), Sequelize ORM, Socket.IO, Groq AI, Brevo API, pdf-parse

- Built an inventory replenishment platform with 42 REST endpoints and Sequelize ORM transactions (BEGIN / COMMIT / ROLLBACK), guaranteeing data integrity during multi-item stock updates.
- Implemented an automated replenishment engine calculating sales velocity and reorder thresholds, paired with an AI document parser evaluating supplier quotation PDFs (pdf-parse).
- Designed Human-in-the-Loop workflows auto-dispatching purchase orders under \$1,000 and holding orders exceeding \$1,000 for manager review under JWT RBAC with instant Brevo API notifications.

Velocity Retail — Fast-Mover Detection & Recommendation Engine (Capstone) Live Demo | GitHub

Node.js, Express, TypeScript, MongoDB, Mongoose, React 18, Redux Toolkit, Tailwind CSS, Socket.IO, Groq AI, Docker Compose, CI/CD

- Architected a retail analytics system identifying fast-moving inventory items using a statistical outlier model (sales velocity exceeding category average by 0.75 standard deviations), reducing overstock risks by 25%.
- Integrated AI inference to generate contextual merchandising recommendations for adjacent in-stock items, broadcasting live stock updates across dashboards via Socket.IO.
- Containerized frontend and backend services using Docker and built automated GitHub Actions CI/CD workflows for type-checking, testing, and deployment.

InsureClaim AI — Medical Claims Adjudication Platform Live Demo

Domo Custom App Architecture (domo.js), React 18, Domo AppDB, Cloud Workflows, AI Document Inspection, Recharts

- Built a medical insurance claim processing dashboard deployed on Domo Cloud, reducing manual audit turnaround from 3 days to real-time automated verification.
- Integrated multimodal AI document checking to audit medical invoices, cross-reference incident chronologies, and flag billing discrepancies.
- Automated claim lifecycle transitions via Domo Cloud Workflows mutating AppDB database records with role-based access security.

Secure Mini Team Task Manager (RBAC & Dual-DB Architecture) Live Demo | GitHub

Node.js, Express.js, Dual Database (MySQL + MongoDB), React, Vite, JWT Refresh Tokens, RBAC, EventEmitter

- Engineered a dual-database task platform: MySQL connection pooling for relational task assignments and MongoDB aggregation pipelines for administrative metrics.
- Secured all endpoints with stateless dual-token JWT (15-min access / 7-day refresh token), password hashing with bcrypt, and Role-Based Access Control (ADMIN, USER).
- Integrated decoupled audit logging using Node.js EventEmitter to record asynchronous security events and task mutations with zero blocking on primary database transactions.

ADDITIONAL PROJECTS & LABS

- **Student Management REST API & Interactive OpenAPI Docs:** Node.js/Express API deployed on Render with interactive Swagger documentation, Joi validation, and Winston logging. Live API Docs | GitHub
- **Aura Workspace Productivity Suite:** React 19 glassmorphism app with Kanban workflow, 3-tier time planner, notes organizer, and DOM virtualization. Live Demo | GitHub
- **Personal Diary & Journaling Web App:** Authenticated private journaling platform built with React and Supabase (PostgreSQL with Row-Level Security). Live Demo | GitHub
- **Product Showcase Landing Page:** Modern, mobile-first product showcase with smooth scroll animations (AOS) and dynamic telemetry counters. Live Demo | GitHub

PROBLEM SOLVING & COMPETITIVE PROGRAMMING

- **LeetCode (50+ Solved):** Actively solving algorithmic challenges across Arrays, Two Pointers, Sliding Window, HashMaps, Trees, Graphs, and Dynamic Programming with regular practice on the **Blind 75** list.
- **HackerRank (30+ Solved):** Earned badges in Algorithmic Optimization, SQL Query Optimization, and Core Programming Paradigms.

EDUCATION

Bachelor of Computer Applications (BCA) — 80% 2022 – 2025

Raja College of Arts and Science (Alagappa University), Ramanathapuram, Tamil Nadu

CERTIFICATIONS

Full Stack Web Development Certification — Whytap Institute